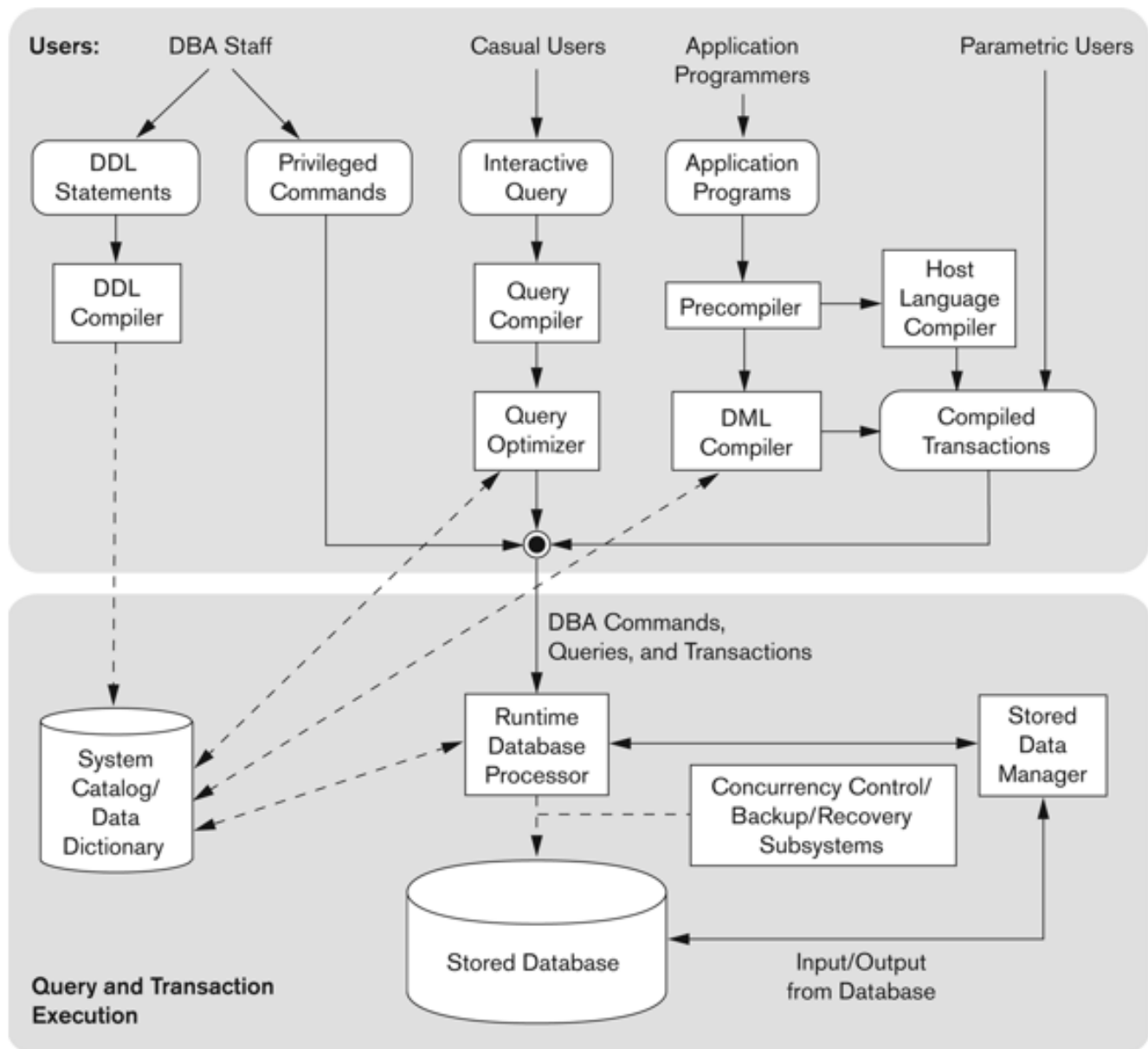


Typical Database Management System Architecture [Diagram]



Typical Database Management System Architecture [Diagram Explanation]

This is a **Typical Database Management System (DBMS) Architecture**, showing how different types of users interact with a database.

Main Sections of the Diagram

The diagram consists of two major sections:

1. **Users and Their Interactions with the DBMS**
 2. **Query and Transaction Execution**
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1. Users and Their Interactions with the DBMS

Different types of users interact with the database system in different ways:

a) DBA Staff (Database Administrators)

- **DDL Statements:** The DBA (Database Administrator) defines the database schema using **Data Definition Language (DDL)**. These statements specify:
 - Table structures
 - Indexes
 - Constraints, etc.
- **DDL Compiler:** It processes the DDL statements and updates the **System Catalog/Data Dictionary**, which stores metadata about the database.
- **Privileged Commands:** DBA staff can also execute special commands for managing security, user access, and performance tuning.

b) Casual Users

- These users interact with the database through **interactive queries**.
- The process:
 - **Query Compiler:** Parses and converts user queries into an internal form.
 - **Query Optimizer:** Optimizes the query for efficient execution.

c) Application Programmers

- **Application Programs:** Developers write applications that interact with the database.
- **Precompiler:** Converts embedded queries in a high-level programming language (e.g., C++, Java) into database commands.
- **DML Compiler:** Processes **Data Manipulation Language (DML)** statements (like SELECT, INSERT, UPDATE, DELETE).
- **Host Language Compiler:** Compiles the application along with the processed database queries.
- **Compiled Transactions:** The final compiled code executes database transactions.

d) Parametric Users

- These users run predefined, frequently used transactions (e.g., in banking or airline reservation systems).
- **Stored Data Manager:** Manages and executes these compiled transactions efficiently.

2. Query and Transaction Execution

Once the user inputs queries or transactions, the DBMS processes them as follows:

1. Query Compilation and Optimization

- The **Query Compiler** translates queries into an intermediate form.
- The **Query Optimizer** improves efficiency by choosing the best execution plan.

2. Runtime Database Processing

- The **Runtime Database Processor** is responsible for actually executing queries and transactions.

3. System Catalog/Data Dictionary

- Stores metadata like table structures, indexes, constraints, and user permissions.
- The DBMS references this information during query execution.

4. Concurrency Control, Backup, and Recovery

- Ensures data integrity when multiple users access the database simultaneously.
- Handles database recovery in case of system failures.

5. **Stored Database and Stored Data Manager**

- The actual **Stored Database** contains the user data.
- The **Stored Data Manager** is responsible for retrieving and storing data efficiently.

Summary of the Flow

1. Users interact with the database through different interfaces.
2. The queries go through **compilation, optimization, and execution**.
3. The **Runtime Database Processor** processes queries and transactions.
4. Data retrieval and storage are managed by the **Stored Data Manager**.
5. **Concurrency control and backup mechanisms** ensure data integrity.

Final Thoughts

This architecture ensures that databases are **efficient, secure, and reliable** for all types of users.